

Muhammad Ali Zaib

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Education

FAST National University of Computer & Emerging Sciences (NUCES)

Aug 2022 – Present

BSc. Data Science

Relevant Courses: Artificial Intelligence, Deep Learning for Perception, Digital Image Processing, Data Mining, MLOps, Agentic AI, Parallel & Distributed Computing

Forman Christian College (A Chartered University)

Aug 2019 – May 2021

FSc Pre-Engineering

Final Year Project

DyslexAI — AI-Powered Literacy Support Platform

GitHub

- Engineered a three-stage OCR pipeline integrating DocTR, TrOCR, and LLM-based post-correction for robust handwriting recognition in dyslexic learners.
- Designed adaptive learning algorithms using real-time performance signals for personalized exercise generation and dynamic skill progression.
- Developed a full-stack platform (FastAPI, React, SQLAlchemy, Supabase) with role-based workflows and a progress analytics dashboard.

Projects

Capability-Aware Verification — Heterogeneous Multi-Agent LLM Framework

GitHub

- Designed a reliability-weighted aggregation framework (CAV) estimating per-agent capability from calibration quality (accuracy, ECE, confidence gap) and paraphrase consistency.
- Benchmarked four strategies across GSM8K, MMLU, and TruthfulQA; CAV outperformed majority voting on MMLU (90.00% vs. 89.25%) and achieved a higher overall average (89.08% vs. 88.92%).
- Ablation studies confirmed the two capability signals are complementary: full CAV reached 91.5% on MMLU vs. 84.0% (calibration-only) and 83.5% (consistency-only).

Fraud Detection MLOps Platform with Adaptive Drift Monitoring

GitHub

- Built an end-to-end fraud detection pipeline on 284K transactions using a weighted XGBoost/LightGBM/RF ensemble with imbalance handling (SMOTE, class-weighting) to maximize recall on the IEEE CIS dataset.
- Engineered a weighted ensemble drift detector (KS-test + PSI + ADWIN) that dynamically penalizes high false-positive detectors, reducing unnecessary retraining triggers.
- Automated the full retraining lifecycle via GitHub Actions CI/CD driven by live production drift signals; monitored via Prometheus and Grafana dashboards with MLflow experiment tracking.

ArtSight — Art Style Classification & Neural Style Transfer

GitHub

- Trained a MobileNetV2 transfer learning model across 10 art style categories for robust multi-class classification.
- Integrated TensorFlow Hub's Arbitrary Neural Style Transfer pipeline for real-time artistic stylization.
- Built a Flask web app supporting custom image uploads and interactive style transformation.

Electric Load Forecasting — Clustering & Predictive Modelling

GitHub

- Built an end-to-end ML pipeline forecasting hourly electricity demand across 10 U.S. cities using Random Forest and XGBoost on weather and consumption data.
- Applied K-Means, Hierarchical Clustering, and PCA for pattern analysis, deployed via an interactive Flask web app with dynamic visualizations.

Skills

Languages Python, C++, SQL

Libraries Pandas, NumPy, Matplotlib, Scikit-learn, PyTorch, TensorFlow, Seaborn, Librosa

Frameworks & Tools FastAPI, Flask, React, Hadoop, Tableau

MLOps & Infra Docker, MLflow, Kubeflow, Git, GitHub

Databases PostgreSQL, Supabase, SQLAlchemy

Machine Learning Supervised/Unsupervised Learning, Deep Learning, NLP, Computer Vision, Transfer Learning

Certifications

Supervised Machine Learning — DeepLearning.ai, Coursera

Verify Jan 2025

Advanced Learning Algorithms — DeepLearning.ai, Coursera

Verify Mar 2025